



2010 Half-Yearly Examination

FORM V

MATHEMATICS EXTENSION 1

Friday 7th May 2010

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators may be used.
- All necessary working should be shown in every question.
- Start each question on a new leaflet.

Structure of the paper

- Total marks — 96
- All eight questions may be attempted.
- All eight questions are of equal value.

Collection

- Write your name, class and master clearly on each leaflet.
- Hand in the eight questions in a single well-ordered pile.
- Hand in a leaflet for each question, even if it has not been attempted.
- If you use a second leaflet for a question, place it inside the first.
- Write your name on the question paper and place it inside your leaflet for Question One.

5A: DNW
5D: MLS
5G: KWM

5B: DS
5E: RCF
5H: REP

5C: TCW
5F: PKH
5I: SJE

Checklist

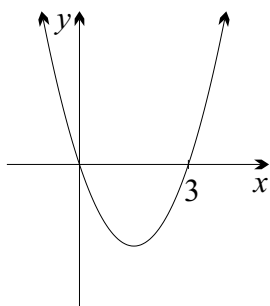
- Writing leaflets: 8 per boy.
- Candidature — 149 boys

Examiner

KWM

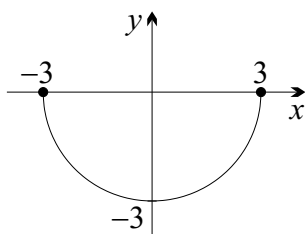
QUESTION ONE (12 marks) Start a new leaflet.

- (a) Write down the exact value of $\cos 150^\circ$.
- (b) Solve $|x + 1| = 3$.
- (c) Consider the straight line $\sqrt{3}x - y + 2 = 0$.
- (i) Write down the gradient of the line.
- (ii) Find the angle of inclination of the line.
- (d) Write down the value of $8^{-\frac{2}{3}}$.
- (e) Factorise $x^3 - 8$.
- (f)



The diagram above shows the curve $y = x(x - 3)$.
Use the curve to solve the inequation $x(x - 3) \geq 0$.

(g)



The diagram above shows the graph of the relation $y = -\sqrt{9 - x^2}$.

- (i) Write down the domain of the relation.
- (ii) Explain why the relation is a function.
- (h) Calculate the limiting sum of the geometric series $3 + 1 + \frac{1}{3} + \dots$.

QUESTION TWO (12 marks) Start a new leaflet.

(a) The first term of an arithmetic progression is -1 and the third term term is 9 .

(i) Find the common difference.

(ii) Find the fiftieth term.

(iii) Find the sum of the first fifty terms.

(b) By rationalising the denominator, express $\frac{3}{2 + \sqrt{5}}$ in the form $a + b\sqrt{5}$.

(c) The function $f(x)$ is defined by $f(x) = x^2 - x$.

(i) Find $f(x + h)$.

(ii) Use the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$ to find $f'(x)$.

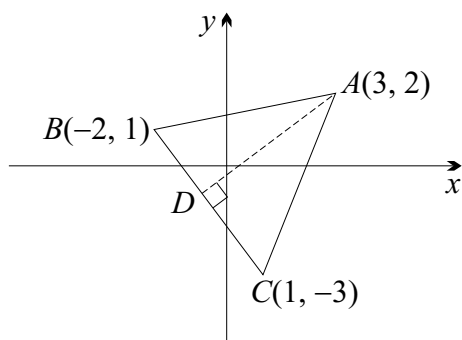
(iii) Find the equation of the tangent to the curve $y = x^2 - x$ at the point $P(1, 0)$.

(d) Solve for x

$$2 \log_a 2 + \log_a 3 = \log_a x.$$

QUESTION THREE (12 marks) Start a new leaflet.

(a)

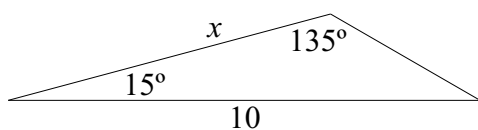


In the diagram above $\triangle ABC$ has vertices $A(3, 2)$, $B(-2, 1)$ and $C(1, -3)$. AD is an altitude of the triangle.

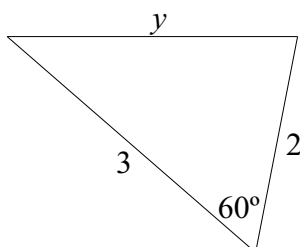
- (i) Find the length of the side BC .
- (ii) Find the gradient of BC .
- (iii) Show that the equation of BC is $4x + 3y + 5 = 0$.
- (iv) Use the perpendicular distance formula to find the length of the altitude AD .
- (v) Hence or otherwise, calculate the area of $\triangle ABC$.

(b) Find the exact value of the pronumeral in each diagram below.

(i)



(ii)



(c) The sum of the first n terms of a sequence is given by $S_n = n^2 - n$.

- (i) Find S_8 and S_9 .
- (ii) Hence find the ninth term of the sequence.

QUESTION FOUR (12 marks) Start a new leaflet.

(a) For each of the following functions find $\frac{dy}{dx}$.

(i) $y = x^3 + 2x - 3$

(ii) $y = (3x + 2)^4$

(iii) $y = \sqrt{x}$

(iv) $y = \frac{3x - 2}{x}$

(b) Use the quotient rule to differentiate $\frac{2x}{x^2 + 1}$.

(c) Use the product rule to differentiate $(2x + 1)^2(x^2 + 1)$ and express the derivative in factored form.

(d) (i) Fully factorise $x^3 - x^2 - 2x$.

(ii) Sketch the curve $y = x^3 - x^2 - 2x$, indicating all intercepts with the axes.

(iii) Hence or otherwise, solve the inequation $x^3 - x^2 - 2x < 0$.

QUESTION FIVE (12 marks) Start a new leaflet.

(a) Given $\tan \theta = -\frac{5}{12}$ and $\cos \theta > 0$, find the exact value of $\sin \theta$.

(b) Sketch the graph of the function $y = 1 - |x - 2|$.

(c) Find the equation of the straight line that passes through the point of intersection of the lines $2x - y + 1 = 0$ and $x + y - 2 = 0$ and also passes through the point $(2, -1)$. Do not find the point of intersection of the two lines and leave your answer in general form.

(d) A function $h(x)$ is defined by $h(x) = \frac{2}{x^2 + 1}$.

(i) Evaluate $h(0)$.

(ii) Show that $h(x)$ is an even function.

(iii) What value does $h(x)$ approach as $x \rightarrow \infty$?

(iv) Sketch the function $y = h(x)$ and state its range.

QUESTION SIX (12 marks) Start a new leaflet.

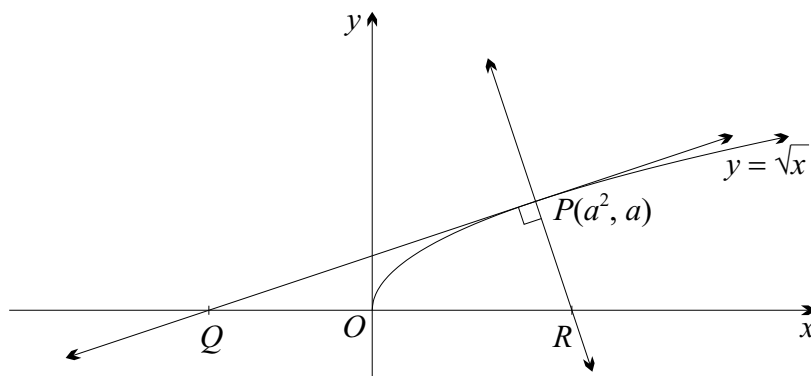
- (a) (i) Solve $\sin x + \sqrt{3} \cos x = 0$, for $0^\circ \leq x \leq 360^\circ$.
(ii) Solve $\sec^2 x + \tan x = 3$, for $0^\circ \leq x \leq 360^\circ$.
(Give your solutions correct to the nearest minute.)
- (b) (i) Write down the radius and centre of the circle $(x+2)^2 + (y-3)^2 = 25$.
(ii) Show that the line $4x - 3y - 8 = 0$ is a tangent to the circle $(x+2)^2 + (y-3)^2 = 25$.
- (c) Prove the identity $\frac{\cos \alpha}{1 + \sin \alpha} + \frac{1 + \sin \alpha}{\cos \alpha} = 2 \sec \alpha$.

QUESTION SEVEN (12 marks) Start a new leaflet.

- (a) The third and sixth terms of a geometric sequence are $\frac{1}{27}$ and $\frac{1}{729}$ respectively.
(i) Find the first term and the common ratio.
(ii) Find the sum of the first seven terms. (Express your answer as a fraction in simplest terms.)
(iii) How many terms of the sequence exceed $\frac{1}{1\,000\,000}$?
- (b) (i) Show that $x = 60^\circ$ is a solution to the equation $\sin x = \frac{1}{2} \tan x$.
(ii) On the same set of axes sketch the graphs of the functions $y = \sin x$ and $y = \frac{1}{2} \tan x$, for $-180^\circ \leq x \leq 180^\circ$.
(iii) Use your graphs to solve $\sin x \leq \frac{1}{2} \tan x$, for $-90^\circ < x < 90^\circ$.

QUESTION EIGHT (12 marks) Start a new leaflet.

(a)



In the diagram above the tangent and normal to the curve $y = \sqrt{x}$ at a variable point $P(a^2, a)$, where $a > 0$, intersect the x -axis at the points Q and R respectively.

- (i) Find the equation of the tangent at the point $P(a^2, a)$.
 - (ii) Find the co-ordinates of Q and R .
 - (iii) Hence show that the difference in the distances of the points Q and R from the origin is constant.
- (b) The two curves $y = x^2 + ax + b$ and $y = cx - x^2$ share a common tangent at the point $(1, 0)$. Find the values of the constants a , b and c .
- (c) Find the equation of the line which bisects the acute angle between the lines $\ell_1 : 3x - 6y - 10 = 0$ and $\ell_2 : 2x - y - 4 = 0$.

END OF EXAMINATION

QUESTION 1

$$(a) \cos 150^\circ = -\cos 30^\circ$$

$$= -\frac{\sqrt{3}}{2} \checkmark$$

$$(b) |x+1| = 3$$

$$x+1=3 \quad \text{or} \quad x+1=-3$$

$$\underline{x=2} \checkmark \quad \text{or} \quad \underline{x=-4} \checkmark$$

$$(c) \sqrt{3}x - y + 2 = 0$$

$$y = \sqrt{3}x + 2$$

$$(i) \text{ gradient} = \sqrt{3}$$

$$(ii) \tan \alpha = \sqrt{3} \checkmark$$

$$\alpha = 60^\circ$$

angle of inclination is $60^\circ \checkmark$

$$(d) 8^{-\frac{2}{3}} = (2)^{-2}$$

$$= \frac{1}{4} \checkmark$$

$$(e) x^3 - 8 = (x-2)(x^2 + 2x + 4) \checkmark$$

$$(f) x(x-3) \geq 0$$

$$x \leq 0, \quad x \geq 3 \checkmark$$

(g)

$$(i) \text{ Domain: } -3 \leq x \leq 3 \checkmark$$

(ii) the graph satisfies the vertical line test. $\checkmark$

$$(h) 3 + 1 + \frac{1}{3} + \dots$$

$$S_\infty = \frac{a}{1-r}$$

$$S_\infty = \frac{3}{1-\frac{1}{3}} \checkmark$$

$$S_\infty = \frac{3}{\frac{2}{3}}$$

$$S_\infty = \frac{9}{2}$$

$$\underline{S_\infty = 4\frac{1}{2} \checkmark}$$

(12)

QUESTION 2

(a)

$$\left. \begin{array}{l} a = -1 \\ a + 2d = 9 \end{array} \right\} \begin{array}{l} \textcircled{1} \\ \textcircled{2} \end{array}$$

$$\begin{aligned} \text{(i)} \quad 2d &= 10 \\ d &= 5 \checkmark \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad t_n &= a + (n-1)d \\ t_{50} &= -1 + 49 \times 5 \\ t_{50} &= 244 \checkmark \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad S_n &= \frac{n}{2}(a+l) \\ S_{50} &= 25(-1 + 244) \\ S_{50} &= 6075 \checkmark \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \frac{3}{2+\sqrt{5}} \times \frac{2-\sqrt{5}}{2-\sqrt{5}} \checkmark \\ = \frac{6-3\sqrt{5}}{4-5} \\ = -6 + 3\sqrt{5} \checkmark \end{aligned}$$

$$\text{(c)} \quad f(x) = x^2 - x$$

$$\begin{aligned} \text{(i)} \quad f(x+h) &= (x+h)^2 - (x+h) \\ &= x^2 + 2xh + h^2 - x - h \checkmark \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x - h - x^2 + x}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh + h^2 - h}{h} \checkmark \\ &= \lim_{h \rightarrow 0} \frac{h(2x + h - 1)}{h} \end{aligned}$$

$$\begin{aligned} &= \lim_{h \rightarrow 0} 2x - 1 + h \\ &= 2x - 1 \checkmark \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad f'(x) &= 2x - 1 \\ f'(1) &= 1 \\ \text{gradient} &= 1 \checkmark \quad P(1, 0) \\ y - y_1 &= m(x - x_1) \\ y &= 1(x - 1) \\ \text{tangent: } y &= x - 1 \checkmark \end{aligned}$$

$$\text{(d)} \quad 2\log_a 2 + \log_a 3 = \log_a x$$

$$\begin{aligned} \log_a 4 + \log_a 3 &= \log_a x \checkmark \\ \log_a 12 &= \log_a x \end{aligned}$$

$$x = 12 \checkmark$$

12

QUESTION 3

(a) $A(3,2), B(-2,1), C(1,-3)$

(i) $\overline{BC} = \sqrt{(-2-1)^2 + (1--3)^2}$

$$\overline{BC} = \sqrt{9+16}$$

$$= 5 \text{ units } \checkmark$$

$$(ii) \text{ gradient } \overline{BC} = \frac{1--3}{-2-1}$$

$$= -\frac{4}{3} \checkmark$$

(iii) $m = -\frac{4}{3} \quad B(-2,1)$

$y - y_1 = m(x - x_1)$

$y - 1 = -\frac{4}{3}(x + 2)$

$3y - 3 = -4x - 8$

$4x + 3y + 5 = 0 \checkmark$

(iv) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$d = \frac{|12 + 6 + 5|}{\sqrt{16 + 9}} \checkmark$

$d = \frac{23}{5}$

$= 4\frac{3}{5} \text{ units } \checkmark$

$$(v) \text{ Area } \triangle ABC = \frac{1}{2} b h$$

$$= \frac{1}{2} \overline{BC} \times \overline{AD}$$

$= \frac{1}{2} \times 5 \times \frac{23}{5}$

$= \frac{23}{2}$

$= 11\frac{1}{2} \text{ sq units. } \checkmark$

(b)

(i) $\frac{x}{\sin 30^\circ} = \frac{10}{\sin 135^\circ} \checkmark$

$\frac{x}{\frac{1}{2}} = \frac{10}{\frac{1}{\sqrt{2}}}$

$2x = 10\sqrt{2}$

$x = 5\sqrt{2} \text{ units. } \checkmark$

(ii)

$y^2 = 4 + 9 - 2 \times 3 \times \cos 60^\circ \checkmark$

$y^2 = 4 + 9 - 6$

$y = \sqrt{7} \text{ units. } \checkmark$

(c)

$S_n = n^2 - n$

$$(i) S_8 = 64 - 8 \quad S_9 = 81 - 9$$

$$= 56 \checkmark \quad S_9 = 72$$

$$(ii) t_9 = S_9 - S_8$$

$$= 72 - 56$$

$t_9 = 16 \checkmark$

(12)

QUESTION 4

(a)

$$(i) \quad y = x^3 + 2x - 3$$

$$\frac{dy}{dx} = 3x^2 + 2 \quad \checkmark$$

(ii)

$$y = (3x+2)^4$$

$$\frac{dy}{dx} = 4(3x+2)^3 \times 3$$

$$= 12(3x+2)^3 \quad \checkmark$$

(iii)

$$y = \sqrt{x}$$

$$y = x^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2} x^{-1/2} \quad \checkmark$$

$$= \frac{1}{2\sqrt{x}}$$

(iv)

$$y = \frac{3x-2}{x}$$

$$y = 3 - 2x^{-1}$$

$$\frac{dy}{dx} = \frac{2}{x^2} \quad \checkmark$$

(b)

$$f(x) = \frac{2x}{x^2+1}$$

$$f'(x) = \frac{(x^2+1)2 - 2x \times 2x}{(x^2+1)^2} \quad \checkmark$$

$$= \frac{2x^2 + 2 - 4x^2}{(x^2+1)^2}$$

$$= \frac{2 - 2x^2}{(x^2+1)^2} \quad \checkmark$$

$$= \frac{2(1-x)(1+x)}{(x^2+1)^2}$$

$$(c) \quad f(x) = (2x+1)^2(x^2+1)$$

$$f'(x) = 2(2x+1)(x^2+1)2 + (2x+1)^2 \times 2x \quad \checkmark$$

$$= 2(2x+1) \{ 2(x^2+1) + x(2x+1) \}$$

$$= 2(2x+1)(4x^2+x+2) \quad \checkmark$$

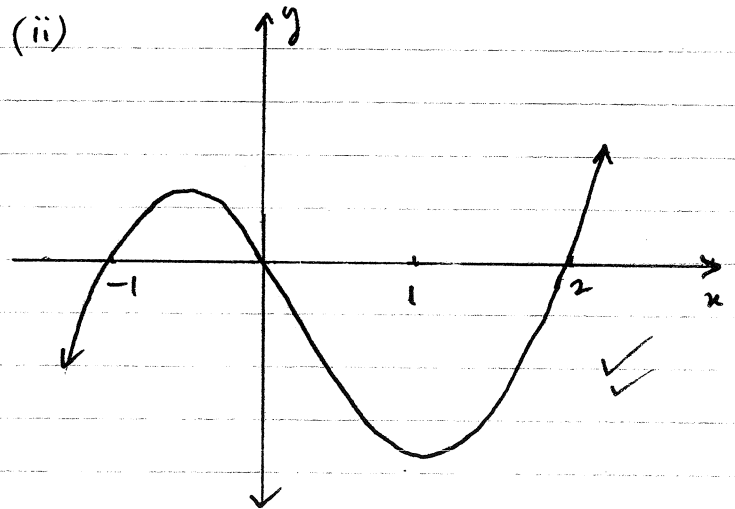
(d) (i)

$$x^3 - x^2 - 2x$$

$$= x(x^2 - x - 2)$$

$$= x(x+1)(x-2) \quad \checkmark$$

(ii)



(iii)

$$x(x+1)(x-2) < 0$$

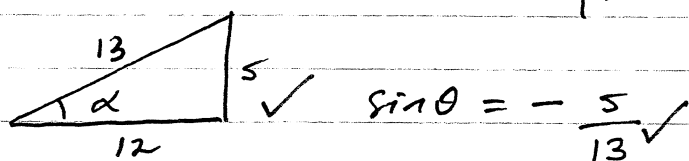
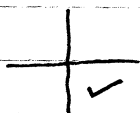
$$x < -1 \quad \text{OR} \quad 0 < x < 2 \quad \checkmark$$

(12)

5.

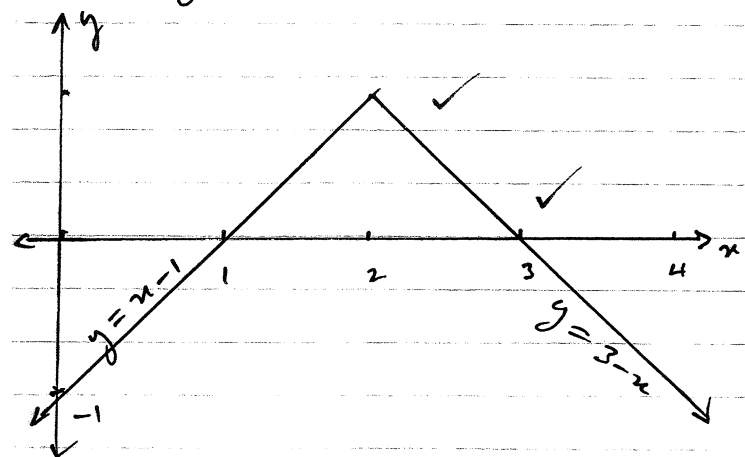
QUESTION 5

(a) $\tan \theta = -\frac{5}{12}$



$\sin \theta = -\frac{5}{13}$

(b) $y = 1 - |x - 2|$



(c)

$2x - y + 1 + k(x + y - 2) = 0$

$(2, -1): 4 + 1 + 1 + k(2 - 1 - 2) = 0$

$6 - k = 0$

$k = 6$

$2x - y + 1 + 6(x + y - 2) = 0$

$8x + 5y - 11 = 0$

(d) $h(x) = \frac{2}{x^2 + 1}$

(i) $h(0) = 2$

(ii) $h(-x) = \frac{2}{(-x)^2 + 1}$

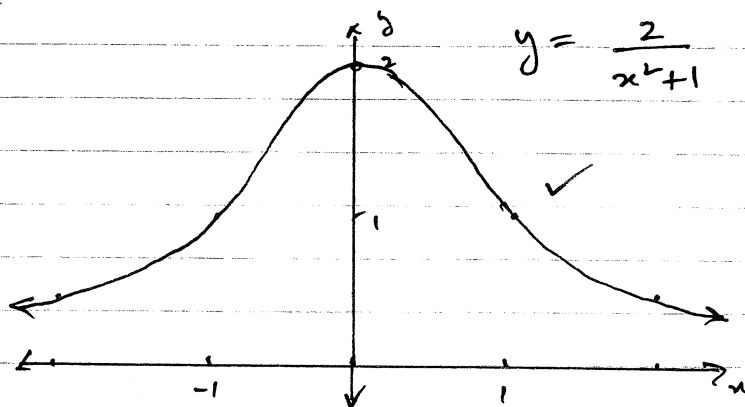
$= \frac{2}{x^2 + 1}$

$= h(x)$

 $\therefore h(x)$ is an even function.

(iii) as $x \rightarrow \infty$, $h(x) \rightarrow 0$

(iv)



Range: $0 < y \leq 2$

12

QUESTION 6

(a)

(i) $\sin x + \sqrt{3} \cos x = 0$

$$\frac{\sin x}{\cos x} + \sqrt{3} = 0$$

$$\tan x = -\sqrt{3} \checkmark$$



$$x = 120^\circ \text{ or } x = 300^\circ$$



(ii) $\sec^2 x + \tan x = 3$

$$(1 + \tan^2 x) + \tan x = 3$$

$$\tan^2 x + \tan x - 2 = 0 \checkmark$$

$$(\tan x - 1)(\tan x + 2) = 0 \checkmark$$

$$\tan x = 1 \quad \tan x = -2$$

$$x = 45^\circ, 225^\circ \quad x = 116^\circ 34', 296^\circ 34' \checkmark$$

(b) (i) $(x+2)^2 + (y-3)^2 = 25$

radius = 5 units

centre $(-2, 3)$

(ii) The distance from the line $4x - 3y - 8 = 0$ to the centre of the circle $(-2, 3)$

$$d = \frac{|-8 - 9 - 8|}{\sqrt{16+9}} \checkmark$$

$$d = \frac{25}{5}$$

$$d = 5 \text{ units.} \checkmark$$

Since the perpendicular distance from the line to the centre equals the radius of the circle, the line intersects the circle at one point only.

Hence the line is a tangent.

(c)

$$\text{LHS} = \frac{\cos \alpha}{1 + \sin \alpha} + \frac{1 + \sin \alpha}{\cos \alpha}$$

$$= \frac{\cos^2 \alpha + (1 + \sin \alpha)^2}{\cos \alpha (1 + \sin \alpha)} \checkmark$$

$$= \frac{\cos^2 \alpha + \sin^2 \alpha + 2 \sin \alpha + 1}{\cos \alpha (1 + \sin \alpha)}$$

$$= \frac{2 + 2 \sin \alpha}{\cos \alpha (1 + \sin \alpha)} \checkmark$$

$$= \frac{2(1 + \sin \alpha)}{\cos \alpha (1 + \sin \alpha)} \checkmark$$

$$= \frac{2}{\cos \alpha}$$

$$= 2 \sec \alpha$$

$$= \text{RHS} //$$

(12)

QUESTION 7

(a)

(i) $ar^2 = \frac{1}{27}$ ——— ①

$ar^5 = \frac{1}{729}$ ——— ②

② $\frac{ar^5}{ar^2} = \frac{1}{729} \div \frac{1}{27}$ ✓

$r^3 = \frac{27}{729}$

$r = \frac{3}{9}$

$r = \frac{1}{3}$ ✓ $a = \frac{1}{3}$

(ii) $S_n = \frac{a(1-r^n)}{1-r}$

$S_7 = \frac{\frac{1}{3}(1-(\frac{1}{3})^7)}{1-\frac{1}{3}}$ ✓

$S_7 = \frac{1}{2} \left(1 - \frac{1}{2187} \right)$

$= \frac{1}{2} \times \frac{2186}{2187}$

$= \frac{1093}{2187}$ ✓

(iii) $t_n > 10^{-6}$

$ar^{n-1} > 10^{-6}$

$\frac{1}{3} \left(\frac{1}{3} \right)^{n-1} > 10^{-6}$

$\frac{1}{3^n} > \frac{1}{1000000}$ ✓

$3^n < 1000000$

$n \log 3 < \log 1000000$

$n \log 3 < 6$

$n < \frac{6}{\log 3}$ ✓

$n < 12.57...$

$n = 12$ ✓

12 terms of the sequence exceed 10^{-6} .

(b)

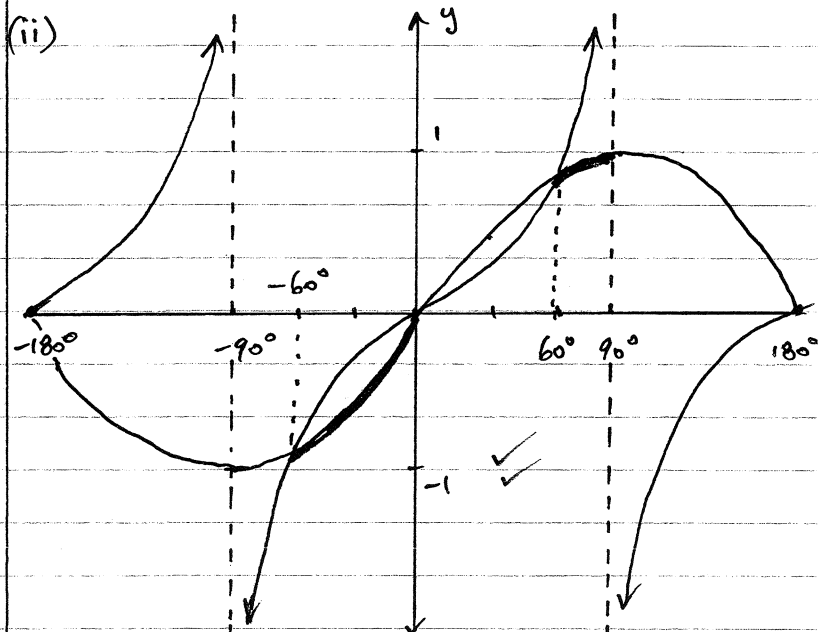
(i) $\sin x = \frac{1}{2} \tan x$

$x = 60^\circ$: LHS = $\sin 60^\circ$
 $= \frac{\sqrt{3}}{2}$

RHS = $\frac{1}{2} \tan 60^\circ$ ✓

$= \frac{1}{2} \times \sqrt{3}$

$= \frac{\sqrt{3}}{2}$ $\therefore 60^\circ$ is a solutⁿ.



(iii) $\sin x \leq \frac{1}{2} \tan x$ $-90^\circ < x < 90^\circ$

$60^\circ \leq x < 90^\circ$ or $-60^\circ \leq x \leq 0^\circ$ ✓

12

QUESTION 8

(a) (i) $y = \sqrt{x}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

at $P(a^2, a)$ gradient $= \frac{1}{2a}$ ✓

$$y - y_1 = m(x - x_1)$$

$$y - a = \frac{1}{2a}(x - a^2)$$

$$2ay - 2a^2 = x - a^2$$

$$x - 2ay + a^2 = 0 \quad \text{tangent.} \quad \checkmark$$

(ii) Q: find $y=0$.

$$x = -a^2$$

$$Q(-a^2, 0)$$

equation of the normal.

$$y - a = -2a(x - a^2) \quad \checkmark$$

put $y=0$

$$-a = -2a(x - a^2)$$

$$\frac{1}{2} = x - a^2$$

$$x = a^2 + \frac{1}{2}$$

$$R(a^2 + \frac{1}{2}, 0) \quad \checkmark$$

(iii) $OR - OQ$

$$|a^2 + \frac{1}{2}| - |-a^2|$$

$$= a^2 + \frac{1}{2} - a^2$$

$$= \frac{1}{2} \quad \checkmark$$

(b) ① $y = x^2 + ax + b$

② $y = cx - x^2$

(1,0) lies on both curves

① $1 + a + b = 0$

② $c - 1 = 0$

$$\therefore c = 1 \quad \checkmark$$

$$\frac{dy}{dx} = 2x + a$$

$$\frac{dy}{dx} = c - 2x$$

at $x=1$ the gradients are equal.

$$a + 2 = c - 2 \quad \checkmark$$

$$a + 2 = 1 - 2$$

$$a + 2 = -1$$

$$a = -3$$

① $1 - 3 + b = 0$

$$b = 2 \quad \checkmark$$

(c)

Let $P(x, y)$ be any point on the angle bisector, d_1 the distance from P to l_1 and d_2 the distance from P to l_2 .

$$\frac{|3x - 6y - 10|}{\sqrt{45}} = \frac{|2x - y - 4|}{\sqrt{5}} \quad \checkmark$$

$$|3x - 6y - 10| = 3|2x - y - 4|$$

$$3x - 6y - 10 = 6x - 3y - 12$$

$$3x + 3y - 2 = 0 \quad \text{or}$$

$$3x - 6y - 10 = -6x + 3y + 12 \quad \checkmark$$

$$9x - 9y - 22 = 0$$

acute

The angle bisector must have a gradient that lies between $l_1: m_1 = \frac{1}{2}$ and $l_2: m_2 = 2$. $\therefore 9x - 9y - 22 = 0$ is ✓
the acute angle bisector of l_1 and l_2 .